

4522 Configuration

Module

Module Options

Lamp Test at Power-Up	No
Protected Start Mode	
ECU Override During Protected Start	
Event Log In Hours Run	
Display SPN Strings	Yes
Enable Fast Loading	No
Maintenance Pin Protected Enable	Yes
Enable Cool Down in Stop Mode	Yes
Power Up in Mode	Stop
All Warnings are Latched	No
Enhanced Tier IV Home Screen	
Show Load Switching Icons	
Limit Audible Alarm Duration	
Enable Text Mode	
Display Previously Active DTCs on GUI	
Log Active DTCs	
Oil Pressure Display Option	Numeric Value
Coolant Temperature Display Option	Numeric Value
Fuel Level Display Option	Numeric Value

Power Saving Options

Backlight Power Save Mode Enable	
Power Save Mode Enable	No
Deep Sleep Mode Enable	
Ignore Active Alarms	

Instrumentation Suppression

Suppress the following instrumentation on the module screen

Generator Frequency	No	Generator Voltage	No
Mains Voltage	No	Mains Frequency	No
Current	No	Power Factor	No
kW	No	kWh	
kVAr	No	kVAh	
kVA	No		

Application

ECU (ECM) Options

Engine Type	[Development] KOEL 4R4K6KSL90 CRS-878
Alternative Engine Speed	No

Inputs**Analogue Input Configuration****ECU (ECM) Options**

Use Module to Measure Oil Pressure	Oil pressure is read from the ECU (ECM)
Use Module to Measure Coolant Temperature	Engine temperature is read from the ECU (ECM)
Use Module to Measure Coolant Level	Engine coolant level is read from the ECU (ECM)

Input Configuration

Analogue Input A	Not Used
Analogue Input B	Not Used
Analogue Input C	Fuel Sensor

'Flexible Analogue' selections are configured on the 'Inputs/Analogue Inputs' pages

'Digital Input' selections are configured on the 'Inputs/Digital Inputs' pages

Oil/Temperature/Fuel/Coolant Level selections are configured on the 'Engine' pages

Analogue Inputs**Analogue Inputs****Flexible Sensor A**

The Analogue Input is currently configured as Not Used

To reconfigure, use the 'Analogue Input Configuration' page

Flexible Sensor C

The Analogue Input is currently configured as a Fuel Sensor

To reconfigure, use the 'Analogue Input Configuration' page

Digital Inputs**Digital Inputs****Digital Inputs****Digital Inputs A - C****Digital Input A**

Function	Emergency Stop
Polarity	Open to Activate
Action	
Arming	
Activation Delay	0s

Digital Input B

Function Remote Start On Load

Polarity Close to Activate

Action

Arming

Activation Delay 0s

Digital Input C

Function User Configured

Polarity Close to Activate

Action Indication

Arming Never

Custom Icon

Activation Delay 0s

Digital Inputs D**Digital Input D**

Function User Configured

Polarity Close to Activate

Action Warning

Arming Always

Custom Icon Monochrome bitmap of size (width x height) 24 x 24 pixels.

Activation Delay 0s

Analogue Inputs**Analogue Input A (Digital)**

The Analogue Input is not configured as a Digital Input
To reconfigure, use the 'Analogue Input Configuration' page

Analogue Input B (Digital)

The Analogue Input is not configured as a Digital Input
To reconfigure, use the 'Analogue Input Configuration' page

Analogue Input C (Digital)

The Analogue Input is not configured as a Digital Input
To reconfigure, use the 'Analogue Input Configuration' page

Digital Outputs

Digital Outputs

	Source	Polarity
Output A	Fuel Relay	Energise
Output B	Start Relay	Energise
Output C	Close Mains Output	Energise
Output D	Close Gen Output	Energise
Output E	Not Used	Energise
Output F	Common Shutdown	Energise

Timers

Timers

Timers

Start Timers

Start Timers

Mains Transient Delay	2s
Start Delay	3s
Engage Attempt	2.0s
Engage Rest	1.5s
Delay Crank	0.1s
Max Start Pause Time	90m 0.1s
Cranking	5.0 s
Cranking Rest	10s
Smoke Limiting	0s
Smoke Limiting Off	0s
DPF Ramp	5.0s
Safety On Delay	10s
Warming	20s
Fuel Pull In Coil Duration	1.0s
ECU Override	10m
Protected Start	30s

Load/Stopping Timers

Load Control Timers

Transfer Delay	0.2s
Breaker Trip Pulse	0.1s
Breaker Close Pulse	0.1s

Load Delay Timers

Delay Load Output 1	0s
Delay Load Output 2	0s
Delay Load Output 3	0s
Delay Load Output 4	0s

Stopping Timers

Return Delay	10s
Cooling	30s
Cooling at Idle	0s
ETS Solenoid Hold	0s
Fail to Stop Delay	30s
Generator Transient Delay	8.0s

Module Timers**Module Timers**

Backlight Power Save Mode Delay	1m
Power Save Mode Delay	1m
Deep Sleep Delay	1m 30s
Page Delay	30s
Audible Alarm	20s

Generator**Generator Options****Alternator**

Alternator Fitted Yes

Poles 4

AC System

AC System 3 Phase, 4 Wire

Generator Voltage**Under Voltage Alarms**

Alarm Yes

Trip 355V PhPh 85.4%

Pre-Alarm Yes

Trip 374V PhPh 90.0%

Loading Voltage

381V PhPh 91.7%

Nominal Voltage

415V PhPh

Over Voltage Alarms

Pre-Alarm Yes

Return 433V PhPh 104.2%

Trip 439V PhPh 105.8%

Shutdown

Trip 460V PhPh 110.8%

Generator Frequency**Under Frequency Alarms**

Alarm Yes

Trip 44.0 Hz 88.0%

Pre-Alarm Yes

Trip 45.0 Hz 90.0%

Loading Frequency

48.0 Hz 96.0%

Nominal Frequency

50.0 Hz	100 %
---------	-------

Over Frequency Alarms

Pre-Alarm Yes

Return 54.0 Hz 108.0%

Trip 55.0 Hz 110.0%

Shutdown Yes

Trip 56.0 Hz 112.0%

Generator Current**Generator Current Options**

Enable CT Support

CT Primary (L1,L2,L3,N) 50 A

CT Secondary 5 Amp

Full Load Rating 34 A

CT Location Gen

Overcurrent Alarm

Immediate Warning Yes

Delayed Alarm Yes

Delayed Alarm Action Shutdown

Delay 1m

Trip 115 % 39 A

Generator Rating

kW Rating 20kW

Overload Protection

Enable Yes

Action Shutdown

Trip 112 % 22 kW

Return 90 % 18 kW

Delay 4s

Mains**Mains Options****Mains Options**

Mains Failure Detection	Yes
Immediate Mains Dropout	No
AC System	3 Phase, 4 Wire

Mains Alarms**Voltage Alarms**

Under Voltage	Yes
Trip	355V PhPh
Return	374V PhPh
Over Voltage	Yes
Return	426V PhPh
Trip	431V PhPh

Frequency Alarms

Under Frequency	Yes
Trip	45.0 Hz
Return	47.0 Hz
Over Frequency	Yes
Return	53.0 Hz
Trip	55.0 Hz

Engine

Engine Protection

Water In Fuel

Action Warning
 Arming Always
 Activation Delay 0s

Fuel Tank Bund

Action Warning

Fan Speed Low

Action Warning
 Arming Always
 Activation Delay 0s

Oil Pressure

Input Type

Oil pressure is read from the ECU (ECM)

Sensor Open Circuit Alarm

Enable Alarm Open circuit alarm applies to the ECU (ECM) reading

Low Oil Pressure Alarms

Enable
 Trip 0.99 Bar 99 kPa, 14.36 PSI

Engine Coolant

Engine Coolant

Engine
 Coolant

Coolant Temperature Alarms

Input Type

Engine coolant level is read from the ECU (ECM)

Sensor Open Circuit Alarm

Enable Alarm Open circuit alarm applies to the ECU (ECM) reading

High Coolant Temperature Alarms

Pre-Alarm
 Return 125 °C 257 °F
 Trip 150 °C 302 °F

Shutdown
Trip 200 °C 392 °F

Coolant Temperature Control**Coolant Heater Control**

Enable No
On 50 °C
Off 55 °C

Coolant Cooler Control

Enable No
On 75 °C
Off 70 °C
Disable when set not available No

Coolant Level Alarms**Input Type**

Engine coolant level is read from the ECU (ECM)

Sensor Open Circuit Alarm

Enable Alarm Open circuit alarm applies to the ECU (ECM) reading

Coolant Level Alarms

Enable
Action None
Arming Always
Trip 25 %
Return 50 %
Delay 0s

Coolant Level Switch

Enable
Action None
Arming Never
Delay 0s

Fuel Options
Fuel Options**Fuel Level****Input Type**

1010kva		
Low Fuel Level Alarms		
Alarm		
Action	Shutdown	
Trip	10 %	
Delay	0s	
Pre-Alarm		
Action	Warning	
Trip	15 %	
Return	20 %	
Delay	0s	
High Fuel Level Alarms		
Pre-Alarm		
Action	None	
Return	98 %	
Trip	99 %	
Delay	0s	
Alarm		
Action	None	
Trip	100 %	
Delay	0s	
Fuel Level Switch		
Enable		
Action	Shutdown	
Arming	Always	
Delay	0s	
Fuel Usage Alarm		
Fuel Usage Alarm		
Enable	No	
Mode	Standard Mode	
Action	Warning Always Latched	
Running Rate	10 % / Hr	10 % / Hr
Stopped Rate	10 % / Hr	10 % / Hr
Engine Options		

ECU (ECM) Options

Engine Type [Development] KOEL 4R4K6KSL90 CRS-878
 Alternative Engine Speed No

ECU (ECM) Options

Module to Record Engine Hours Yes
 Module To Use Engine Speed No
 Module To Use Charge Alt Voltage Yes
 Disable ECM Speed Control No

Miscellaneous Options

J1939-75 Instrumentation Enable Yes
 J1939-75 Alarms Enable No
 CAN source address (instrumentation) 44

Startup Options

Start Attempts 3

Speed Sensing

Enable Multiple Engage Attempts No
 Engage Attempts 2

Pre-heat

Enabled Yes
 On 50 °C 122 °F
 On 50 °C 122 °F
 Duration 0s

Post-heat

Enabled Yes
 On 50 °C 122 °F
 On 50 °C 50 °C
 Duration 0s

Overspeed Options

Overspeed Overshoot 10 %
 Overshoot Delay 2s

Exhaust Temperature

Exhaust temp. based Cooling 0s
 Trip 650 °C 1202 °F

ECU (ECM) Options**DPF Regeneration Control**

Allow Non-Mission Regeneration No

ECU Wakeup

Enable No

Periodic Wakeup Time 1h

Coolant Measurement Persistence No

SPN Ignore List

SPN FMI

1

2

3

4

5

SPN FMI

6

7

8

9

10

Miscellaneous

CAN source address (engine messages) 234

ECU (ECM) Alarms**ECU (ECM) Data Fail****ECU (ECM) Data Fail**

Action Shutdown

Arming From Safety On

Activation Delay 0s

DM1 Signals**ECU Amber**

Action Warning

Arming Always

Activation Delay 0s

ECU Red

Action Warning
Arming Always
Activation Delay 0s

ECU Malfunction

Action Warning
Arming Always
Activation Delay 0s

ECU Protect

Action Shutdown
Arming Always
Activation Delay 0s

Other Specific Signals**DPTC Filter**

Enabled Yes
Action Warning
Arming From Safety On

HEST Active

Enabled Yes
Action Warning
Arming From Safety On

DEF Level

Enabled Yes
Action Warning
Arming From Safety On
Activation Delay 0s

SCR Inducement

Enabled Yes
Action Warning
Arming From Safety On
Activation Delay 0s

Gas Engine Options**Gas Engine Timers**

Choke 2s

Gas on Delay	2s
Ignition Off Delay	2s

Crank Disconnect**Options**

Crank Disconnect on Oil Pressure	No
Check Oil Pressure Prior to Starting	Yes

Crank Disconnect

Generator Frequency	15.0 Hz
Engine Speed	600 RPM
Oil Pressure	3 Bar
Delay	0s
Charge Alternator	No
6.0 V DC	

Speed Settings**Under Speed Shutdown**

Enable	Yes
1310 RPM	

Over Speed Shutdown

Trip	1710 RPM
------	----------

Plant Battery**Voltage Alarms**

Under Voltage	Yes
Warning	9.5 V DC
Return	10.0 V DC
Delay	10s
Over Voltage	Yes
Return	14.5 V DC
Warning	15.0 V DC
Delay	10s

Charge Alternator Alarm

Shutdown	Enabled
Warning	Enabled

Start On Low Battery

Enable	No
Threshold	18.0 V DC

Engine Run Duration	1h
Start Delay	5s

Communications

Basic

Server ID 2

Baud Rate 9600

Advanced

Client inactivity timeout 5s

Scheduler

Exercise Scheduler

Enabled

No

Scheduled runs are On Load

No

Schedule Period

Weekly

Week	Day	Start Time	Duration
	Monday	00:00	00:00
	Monday	00:00	00:00
	Monday	00:00	00:00
	Monday	00:00	00:00
	Monday	00:00	00:00
	Monday	00:00	00:00
	Monday	00:00	00:00
	Monday	00:00	00:00

Maintenance Alarm**Maintenance Alarm Oil**

Enable	Yes
Action	Warning
Engine Run Hours	500 hrs

Maintenance Alarm Air

Enable	No
Action	Warning
Engine Run Hours	10 hrs

Maintenance Alarm Fuel

Enable	No
Action	Warning
Engine Run Hours	10 hrs

Alternative Configuration**Alternative Configuration Options****Alternative Configuration Options**

Default Configuration	Main Configuration
-----------------------	--------------------

Configuration 1**Configuration Options****Enable Configuration**

Enable Configuration	No
----------------------	----

Generator**Generator Options****AC System**

AC System	3 Phase, 4 Wire
-----------	-----------------

Generator Voltage**Under Voltage Alarms**

Alarm	Yes	
Trip	318V PhPh	80.0%
Pre-Alarm	Yes	
Trip	339V PhPh	85.2%

Loading Voltage

358V PhPh	90.0%
-----------	-------

Nominal Voltage

398V PhPh

Over Voltage Alarms

Pre-Alarm Yes

Return 439V PhPh 110.4%

Trip 458V PhPh 115.2%

Shutdown

Trip 479V PhPh 120.4%

Generator Frequency**Under Frequency Alarms**

Alarm Yes

Trip 40.0 Hz 80.0%

Pre-Alarm Yes

Trip 42.0 Hz 84.0%

Loading Frequency

45.0 Hz 90.0%

Nominal Frequency

50.0 Hz 100 %

Over Frequency Alarms

Pre-Alarm Yes

Return 54.0 Hz 108.0%

Trip 55.0 Hz 110.0%

Shutdown Yes

Trip 57.0 Hz 114.0%

Generator Current**Generator Current Options**

CT Primary (L1,L2,L3) 600 A

CT Secondary 5 Amp

Full Load Rating 500 A

Overcurrent Alarm

Immediate Warning Yes

Delayed Alarm Yes

Delayed Alarm Action Electrical Trip

Delay 1m

Trip 100 % 500 A			
Generator Rating			
kW Rating 200kW			
Overload Protection			
Enable		No	
Action		Shutdown	
Trip	100 %	200	kW
Return	90 %	180	kW
Delay	5s		

Mains
Mains Options**Mains Options**

Mains Failure Detection	Yes
Immediate Mains Dropout	No
AC System	3 Phase, 4 Wire

Mains Alarms**Voltage Alarms**

Under Voltage	Yes
Trip	318V PhPh
Return	358V PhPh
Over Voltage	Yes
Return	438V PhPh
Trip	478V PhPh

Frequency Alarms

Under Frequency	Yes
Trip	45.0 Hz
Return	48.0 Hz
Over Frequency	Yes
Return	52.0 Hz
Trip	55.0 Hz

Engine

Engine Options

ECU (ECM) Options

Engine Type [Development] KOEL 4R4K6KSL90 CRS-878
 Alternative Engine Speed No

Speed Settings

Under Speed Shutdown

Enable No
 1200 RPM

Over Speed Shutdown

Trip 1710 RPM

Configuration 2

Configuration Options

Enable Configuration

Enable Configuration No

Generator

Generator Options

AC System

AC System 3 Phase, 4 Wire

Generator Voltage

Under Voltage Alarms

Alarm Yes
 Trip 318V PhPh 80.0%
 Pre-Alarm Yes
 Trip 339V PhPh 85.2%

Loading Voltage

358V PhPh 90.0%

Nominal Voltage

398V PhPh

Over Voltage Alarms

Pre-Alarm Yes
 Return 439V PhPh 110.4%
 Trip 458V PhPh 115.2%

Shutdown			
Trip	479V PhPh	120.4%	

Generator Frequency

Under Frequency Alarms

Alarm	Yes		
Trip	40.0 Hz	80.0%	
Pre-Alarm	Yes		
Trip	42.0 Hz	84.0%	

Loading Frequency

45.0 Hz	90.0%
---------	-------

Nominal Frequency

50.0 Hz	100 %
---------	-------

Over Frequency Alarms

Pre-Alarm	Yes		
Return	54.0 Hz	108.0%	
Trip	55.0 Hz	110.0%	
Shutdown	Yes		
Trip	57.0 Hz	114.0%	

Generator Current

Generator Current Options

CT Primary (L1,L2,L3)	600 A
CT Secondary	5 Amp
Full Load Rating	500 A

Overcurrent Alarm

Immediate Warning	Yes
Delayed Alarm	Yes
Delayed Alarm Action	Electrical Trip
Delay	1m
Trip	100 % 500 A

Generator Rating

kW Rating	200kW
-----------	-------

Overload Protection

Enable				No
Action				Shutdown
Trip	100 %	200	kW	
Return	90 %	180	kW	
Delay	5s			

Mains
Mains Options**Mains Options**

Mains Failure Detection	Yes
Immediate Mains Dropout	No
AC System	3 Phase, 4 Wire

Mains Alarms**Voltage Alarms**

Under Voltage	Yes
Trip	318V PhPh
Return	358V PhPh
Over Voltage	Yes
Return	438V PhPh
Trip	478V PhPh

Frequency Alarms

Under Frequency	Yes
Trip	45.0 Hz
Return	48.0 Hz
Over Frequency	Yes
Return	52.0 Hz
Trip	55.0 Hz

**Engine
Engine Options****ECU (ECM) Options**

Engine Type [Development] KOEL 4R4K6KSL90 CRS-878
Alternative Engine Speed No

Speed Settings**Under Speed Shutdown**

Enable No
1200 RPM

Over Speed Shutdown

Trip 1710 RPM

**Configuration 3
Configuration Options****Enable Configuration**

Enable Configuration No

**Generator
Generator Options****AC System**

AC System 3 Phase, 4 Wire

Generator Voltage**Under Voltage Alarms**

Alarm Yes
Trip 318V PhPh 80.0%
Pre-Alarm Yes
Trip 339V PhPh 85.2%

Loading Voltage

358V PhPh 90.0%

Nominal Voltage

398V PhPh

Over Voltage Alarms

Pre-Alarm Yes
Return 439V PhPh 110.4%
Trip 458V PhPh 115.2%

Shutdown			
	Trip	479V PhPh	120.4%

Generator Frequency**Under Frequency Alarms**

Alarm	Yes		
	Trip	40.0 Hz	80.0%
Pre-Alarm	Yes		
	Trip	42.0 Hz	84.0%

Loading Frequency

45.0 Hz	90.0%
---------	-------

Nominal Frequency

50.0 Hz	100 %
---------	-------

Over Frequency Alarms

Pre-Alarm	Yes		
	Return	54.0 Hz	108.0%
	Trip	55.0 Hz	110.0%
Shutdown	Yes		
	Trip	57.0 Hz	114.0%

Generator Current**Generator Current Options**

CT Primary (L1,L2,L3)	600 A
CT Secondary	5 Amp
Full Load Rating	500 A

Overcurrent Alarm

Immediate Warning	Yes
Delayed Alarm	Yes
Delayed Alarm Action	Electrical Trip
Delay	1m
	Trip 100 % 500 A

Generator Rating

kW Rating	200kW
-----------	-------

Overload Protection

Enable				No
Action				Shutdown
Trip	100 %	200	kW	
Return	90 %	180	kW	
Delay	5s			

Mains
Mains Options**Mains Options**

Mains Failure Detection	Yes
Immediate Mains Dropout	No
AC System	3 Phase, 4 Wire

Mains Alarms**Voltage Alarms**

Under Voltage	Yes
Trip	318V PhPh
Return	358V PhPh
Over Voltage	Yes
Return	438V PhPh
Trip	478V PhPh

Frequency Alarms

Under Frequency	Yes
Trip	45.0 Hz
Return	48.0 Hz
Over Frequency	Yes
Return	52.0 Hz
Trip	55.0 Hz

ECU (ECM) Options

Alternative Engine Speed

No

Under Speed Shutdown

1200 RPM

Trip 1710 RPM

Advanced

Instrumentation Configuration

Description

10 Configurable CAN 10

Instrumentation Configuration

	Enabled	Source
1		<Not Used>
2		<Not Used>
3		<Not Used>
4		<Not Used>
5		<Not Used>